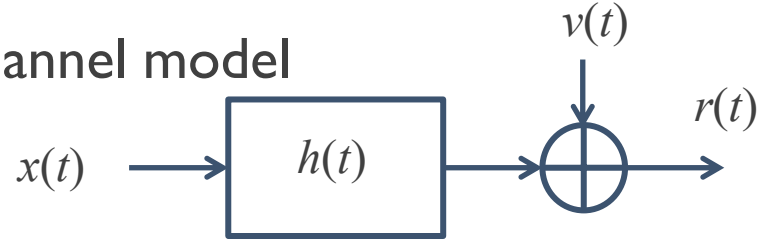
Lecture #14

EE 471C / EE 381K-17 Wireless Communication Lab

From previous lectures

- Frequency selective channel model



- The channel includes all **frequency selective effects**
  - Pulse-shaping filter
  - Transmit signal scaling
  - Propagation effects
  - Receive filter
- **Previous lectures**: channel estimation and equalization
- **This lecture**: frequency offset correction

Preview of today´s lecture

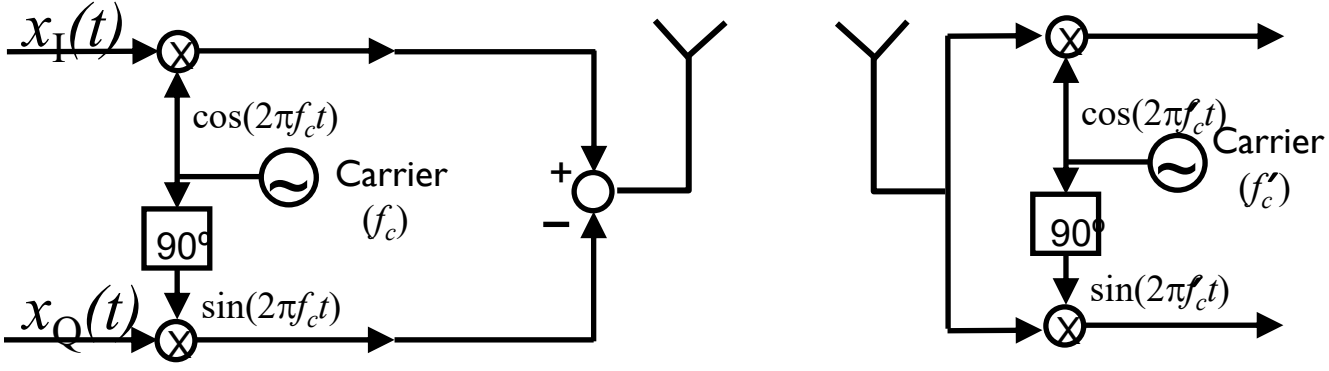
- Carrier frequency offset correction in frequency-selective channels
  - Model for frequency offset in frequency-selective channels
  - Application of single frequency estimation
- Frequency offset estimation and frame synchronization using periodic training for single carrier systems
- Frequency offset estimation and frame synchronization using periodic training for OFDM

Model for frequency offset in frequency-selective channels

Learning objectives

- Understand the received signal model that includes frequency offset

Origin of frequency offset



- Consider the received signal after downconversion
- If only f_c' not f_c is known at the receiver?
  - The result is carrier frequency offset (CFO) $f_e = f_c' - f_c$

$$r'(t) = e^{\mathrm{j}2\pi f_e t}\left(r_\mathrm{i}(t) + \mathrm{j}r_\mathrm{q}(t)\right)$$
$$= e^{\mathrm{j}2\pi f_e t} r(t).$$

Signal model including frequency offset

- Carrier frequency offset multiplies the matched filtered received signal by $e^{\mathrm{j}2\pi f_{\mathrm{e}} t}$

$$y(t) = e^{\mathrm{j}2\pi f_{\mathrm{e}} t} \int r(t-\tau) g_{\mathrm{rx}}(\tau) \mathrm{d}\tau.$$

- Sampling at the symbol rate and using the FIR channel model

$$y[n] = e^{\mathrm{j}2\pi \epsilon n} \sum_{\ell=0}^{L} h[\ell] s[n-\ell] + v[n]$$

- Correction of frequency offset
  - Estimate ϵ and derotate the received signals

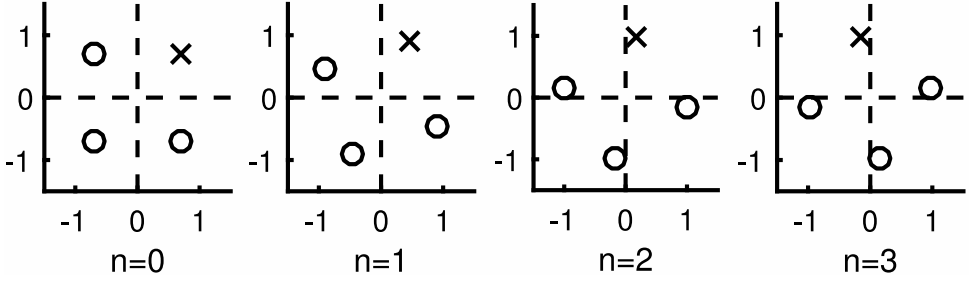
$$e^{-\mathrm{j}2\pi \epsilon n} y[n]$$

Frequency offset in frequency selective channels

- Frequency flat case
  - Successive rotation of each symbol by $\exp(j2\pi\epsilon n)$

$$y[n] = e^{j2\pi\epsilon n} hs[n] + v[n]$$

  - Successive rotation of the constellation



- **Frequency selective case**
  - Rotation applied to convolution of the symbols and the channel

$$y[n] = e^{\mathrm{j}2\pi\epsilon n} \sum_{\ell=0}^{L} h[\ell]s[n-\ell] + v[n]$$

  - Offset happens after the convolution with the channel

Model for frequency offset in frequency-selective channels

Learning objectives

- Understand the received signal model that includes frequency offset

Revisiting single frequency estimation

Learning objectives

- Extend simple frequency offset estimator for flat channels to frequency selective channels

Frequency offset est. in a flat channel using a sinusoid

- Suppose that frame synchronization has been solved
- Choose as a training sequence a complex sinusoid

$$t[n] = \exp(\mathrm{j}\omega n) \; for \; n = 0, 1, \ldots, N_{\mathrm{tr}} - 1$$

- The received signal during training is

$$y[n] = e^{\mathrm{j}2\pi\epsilon n} e^{\mathrm{j}\omega n} h + v[n]$$

- Removing the known offset

$$e^{-\mathrm{j}\omega n} y[n] = e^{\mathrm{j}2\pi\epsilon n} h + e^{-\mathrm{j}\omega n} v[n]$$

This becomes a classic problem of estimating a complex sinusoid with unknown amplitude, frequency and phase (amplitude and phase from the channel)

- The offset can be estimated using LS for example

How does this work for frequency selective channels?

- Now the received signal is

$$y[n] = e^{\mathrm{j}2\pi\epsilon n} \sum_{\ell=0}^{L} h[\ell] t[n-\ell] + v[n]$$

Discarding samples that do not depend on the training (remove edge effects)

$$n = L, L+1, \ldots, N_{\mathrm{tr}} - 1$$

- For the exponential training signal $t[n] = \exp(\mathrm{j}2\pi f_{\mathrm{t}} n)$

$$\begin{aligned} y[n] &= e^{\mathrm{j}2\pi\epsilon n} \sum_{\ell=0}^{L} h[\ell] e^{\mathrm{j}2\pi f_{\mathrm{t}}(n-\ell)} + v[n] \\ &= e^{\mathrm{j}2\pi\epsilon n} e^{\mathrm{j}2\pi f_{\mathrm{t}} n} \sum_{\ell=0}^{L} h[\ell] e^{-\mathrm{j}2\pi f_{\mathrm{t}}\ell} + v[n] \\ &= e^{\mathrm{j}2\pi\epsilon n} e^{\mathrm{j}2\pi f_{\mathrm{t}} n} \mathsf{h}\left(e^{\mathrm{j}2\pi f_{\mathrm{t}}}\right) + v[n] \end{aligned}$$

DTFT of h[n]

Rotation by the known freq. of the training sinusoid

Simple estimator for frequency selective channels

- Derotate by the frequency of the training signal

$$e^{-\mathrm{j}2\pi f_{\mathrm{t}} n} y[n] = e^{\mathrm{j}2\pi \epsilon n} \mathsf{h}\left(e^{\mathrm{j}2\pi f_{\mathrm{t}}}\right) + e^{-\mathrm{j}2\pi f_{\mathrm{t}}} v[n]$$

- Same form as in the flat channel case
- The same estimator can be applied to frequency selective channels!
- This approach was taken in the GSM standard

Revisiting single frequency estimation

Learning objectives

- Extend simple frequency offset estimator for flat channels to frequency selective channels

Frequency offset estimation using periodic training

Learning objectives

- Use the Moose method to perform joint frequency offset estimation and frame synchronization

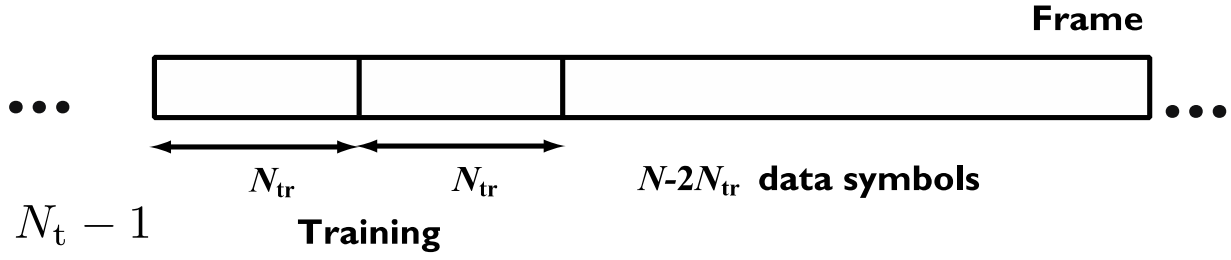
Frequency offset estimation using periodic training

- Focus on a tricky approach for estimating carrier frequency offset using periodic structure in the transmitted signal
  - P. H. Moose, "A technique for orthogonal frequency division multiplexing frequency offset correction," IEEE Trans. on Commun. vol. 42, no.10, pp.2908-2914, Oct 1994.

- This approach was described for OFDM but applies more generally to single carrier as well

- Apply a more sophisticated variation to OFDM later

Periodic training structure



$L \leq n \leq N_t - 1$

- The Moose algorithm exploits periodicity in the training sequence
- Let training start at n=0

$$s[n] = s[n + N_{tr}] = t[n] \qquad n = 0, 1, \ldots, N_{tr} - 1$$

- For $L \leq n \leq N_{tr}$-1

$$y[n] = e^{j2\pi\epsilon n} \sum_{\ell=0}^{L} h[\ell] s[n - \ell] + v[n]$$

$$y[n + N_{tr}] = e^{j2\pi\epsilon(n+N_{tr})} \sum_{\ell=0}^{L} h[\ell] s[n + N_{tr} - l] + v[n + N_{tr}]$$

Exploiting the periodicity

- Note that $s[n+N_{\text{tr}}] = s[n] = t[n]$ for $n = 0, 1, \ldots, N_{\text{tr}} - 1$

$$y[n+N_{\text{tr}}] = e^{\text{j}2\pi\epsilon N_{\text{tr}}} e^{\text{j}2\pi\epsilon n} \sum_{\ell=0}^{L} h[\ell] t[n-\ell] + v[n+N_{\text{tr}}]$$

$$\approx e^{\text{j}2\pi\epsilon N_{\text{tr}}} y[n].$$

It does not depend on the unknown channel coefficients

There is a relationship between different parts of the received signal

- Least squares is one solution but ϵ in the exponent requires a nonlinear least squares solution

Relaxing the offset problem

- A solution is to consider a relaxed problem

$$y[n + N_{\text{tr}}] = ay[n]$$

- Solving for the LS estimate of a and find the offset from the phase

- This problem is similar to flat fading channel estimation

Frequency offset estimate from relaxed solution

- Simple frequency offset estimator

$$\hat{\epsilon}_{\mathrm{LS}} = \frac{1}{2\pi N_{\mathrm{tr}}} \mathrm{phase}\left(\frac{\sum_{n=0}^{N_{\mathrm{tr}}-1} y^*[n] y[n+N_{\mathrm{tr}}]}{\sum_{n=0}^{N_{\mathrm{tr}}-1} y^*[n] y[n]} \right)$$

$$= \frac{1}{2\pi N_{\mathrm{tr}}} \mathrm{phase}\left(\sum_{n=L}^{N_{\mathrm{tr}}-1} y^*[n] y[n+N_{\mathrm{tr}}] \right)$$

Neglect the denominator since it does not contribute to the phase

where phase denotes the principle phase of the argument

- This is also the maximum likelihood estimate

Estimator range depends on the training length

- The estimate of ϵ will only be accurate for $|\epsilon N_{\text{tr}}| \leqslant \frac{1}{2}$ or equivalently

$$|\epsilon| \leqslant \frac{1}{2N_{\text{tr}}}, \quad \text{or} \quad |f_{\text{e}}| \leqslant \frac{1}{2TN_{\text{tr}}}$$

  - Choosing larger N_{tr} improves the estimate but reduces the range of offsets that can be corrected
  - Choosing smaller N_{tr} improves the range but is more susceptible to noise
- Can use multiple shorter periods to achieve high range
  - Use multiple repetitions of a short training sequence

- IEEE 802.11a/n/ac and related standards use a combination of both repeated short training sequences and long training sequences

Example

- *Compute the maximum allowable offset for a* 1Ms/s QAM *signal, with* $f_c = 2$ GHz, *and* $N_{\text{tr}} = 10$

$$\max |f_e| = \frac{1}{2TN_{\text{t}}} = \frac{1}{2}10^6\frac{1}{10} = \frac{1}{2}10^5 = 50\text{kHz}$$

In terms of parts per million, we need an oscillator that can generate a 2 GHz carrier with an accuracy of $50\text{e}3/2\text{e}9 = 25ppm$

Example

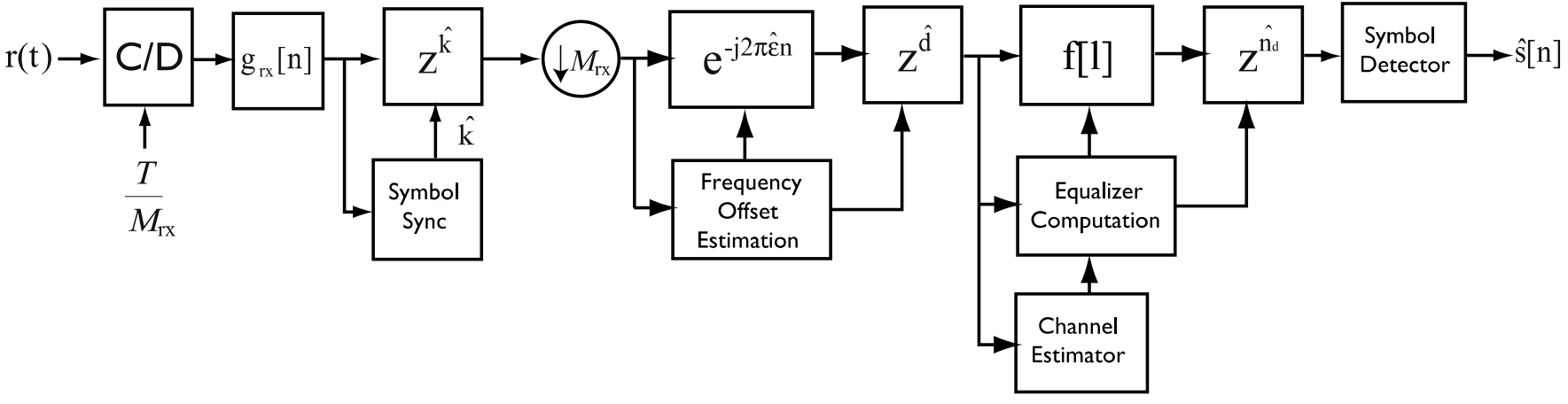
- Consider a wireless system where each data frame is preceded by two training blocks each consisting of N_{tr} =12 training symbols. Let the symbol period be T =4μs. What is the maximum frequency offset that can be corrected using training?

Frame synchronization

- A side benefit of the Moose algorithm is that it also provides a nice way of performing frame synchronization
- The correlation peak should occur when the pair of training sequences is encountered at the receiver
- Solve for the offset d such that

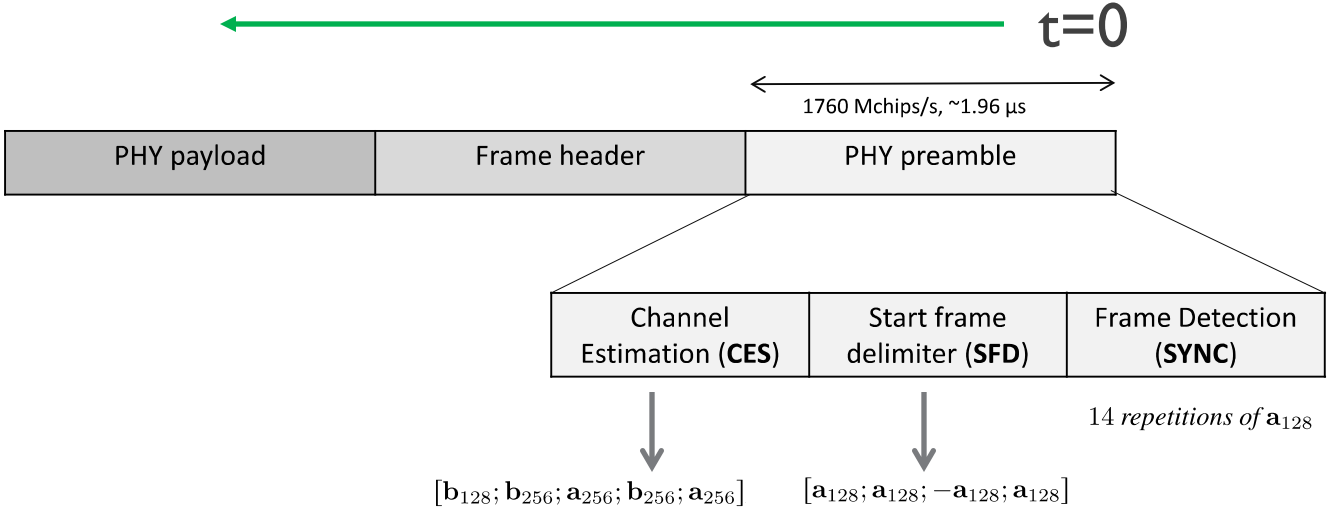
$$\hat{d} = \arg\max \frac{\left|\sum_{n=L}^{N_{\text{tr}}-1} y[n+\Delta+N_{\text{tr}}]y^*[n+\Delta]\right|^2}{\left(\sum_{n=L}^{N_{\text{tr}}-1} |y[n+\Delta]|^2\right)^2}$$

Complete QAM receiver



Example

- *Consider the IEEE 802.15.3c preamble structure for the high rate mode of the SC-PHY. Explain how to use the SYNC and SFD fields for carrier frequency offset estimation and frame synchronization.*



$\{a[n]\}_{n=0}^{N_g-1}$ and $\{b[n]\}_{n=0}^{N_g-1}$ Golay complementary sequences

Using the SYNC field for synchronization

Frame Detection (**SYNC**)

14 *repetitions of* $\mathbf{a}_{128}$

- The maximum supported channel length is 128
- Frame detection algorithm

$$\hat{d} = \arg\max \sum_{p=1}^{13} \frac{\left|\sum_{n=0}^{127} y[n+\Delta+128p+128]y^*[n+\Delta+128p]\right|^2}{\left(\sum_{n=0}^{127} |y[n+\Delta+128+128p]|^2\right)^2}$$

First repetition of the SYNC acts as cyclic prefix

- Frequency offset estimator

$$\hat{\epsilon} = \frac{1}{2\pi 128} \sum_{p=1}^{13} \text{phase}\left(\sum_{n=0}^{127} y[n+\hat{d}+128p+128]y^*[n+\hat{d}+128p]\right)$$

Frequency offset estimation using periodic training

Learning objectives

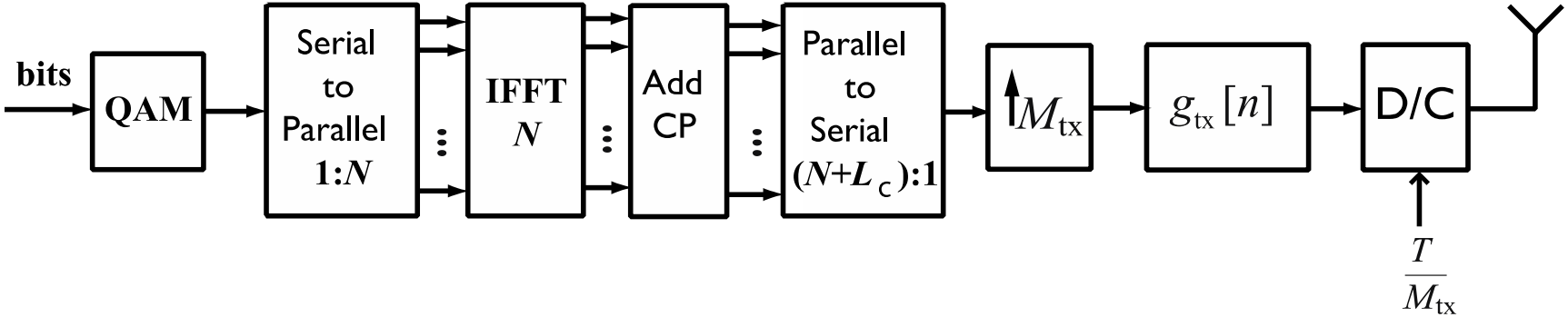
- Use the Moose method to perform joint frequency offset estimation and frame synchronization

Frequency offset estimation in OFDM

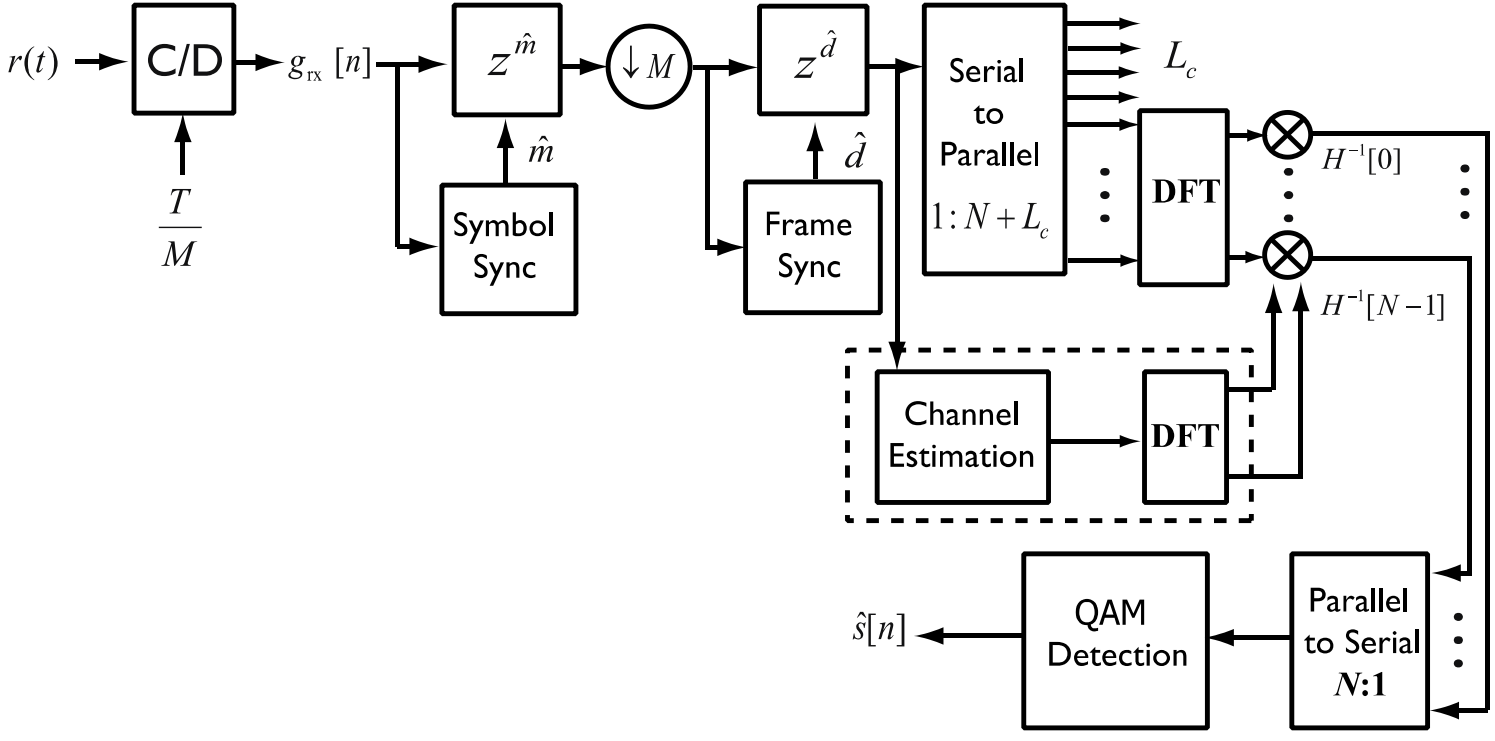
Learning objectives

- Describe and implement carrier frequency offset estimation for an OFDM system

OFDM transmitter



OFDM receiver



Motivation for the Schmidl and Cox algorithm

- Moose's method can be used directly provided that two OFDM symbols are repeated
- It is of interest to develop methods that can work with only a single OFDM symbol
- Modification proposed by Schmidl and Cox
  - Create periodicity in one OFDM symbol instead of repeating multiple OFDM symbols
  - Use a second OFDM symbol to enhance correction range

T. Schmidl and D. Cox, "Robust frequency and timing synchronization for OFDM," IEEE Trans. Commun., 1997

Creating periodicity through zeros

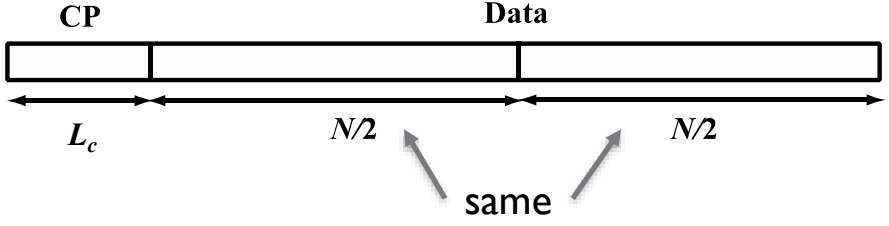
- As an example consider turning off the odd subcarriers

$$w[n] = \frac{1}{N}\sum_{m=0}^{\frac{N}{2}-1} s[2m]e^{j\frac{2\pi 2m(n-L_c)}{N}} = \frac{1}{N}\sum_{m=0}^{\frac{N}{2}-1} s[2m]e^{j\frac{2\pi m(n-L_c)}{N/2}}$$

$$\Rightarrow \text{for } n \geq L_c \text{ and } n < L_c + \frac{N}{2},\ \ w[n] = w\left[n + \frac{N}{2}\right]$$

- This means that the OFDM signal contains a portion which is periodic
  - Apply Moose algorithm

$$w[n + L_c] \ = \ w[n + L_c + N/2] \qquad \text{for } n = 0, 1, \ldots, N/2 - 1$$



Extends to turn off K-1 carriers

⟶ periodicity of N/K

Exploiting periodicity for offset estimation

- Consider the received signal

$$y[n] \;=\; e^{j2\pi\epsilon n}\sum_{\ell=0}^{L} h[\ell]w[n-\ell] + v[n]$$

- Let $\bar{y}[n] = y[n+L_c]$. Because of the periodicity, in the absence of noise

$$\bar{y}[n+N/2] \;=\; e^{j2\pi\epsilon N/2}\bar{y}[n]$$

- Simple frequency offset estimator follows as before

$$\hat{\epsilon}_{\rm LS} = \frac{1}{2\pi N_{\rm tr}}\,\text{phase}\left(\sum_{n=L}^{N/2-1}\bar{y}^*[n]\bar{y}[n+N]\right)$$

Correlation

Performs OFDM symbol synchronization and frequency offset estimation jointly

Frequency offset range

$\hat{\epsilon}$ $\epsilon - \hat{\epsilon}$

- Maximum correctable offset is up to one subcarrier width

$$|\epsilon| \leq \frac{1}{2N/2} = \frac{1}{N}$$

Fine offset correction

  - Unfortunately, in an OFDM system N is quite large

- Decompose the offset as

$$\epsilon = m\frac{1}{N/2} + \epsilon_f$$

Integer offset

Fine offset ⟹ Estimate using correlation

  - Can estimate the **fine offset** using the correlation as described

Selecting a second training symbol

- Let $\{t_1[n]\}_{n=0}^{N/2-1}$ and $\{t_2[n]\}_{n=0}^{N/2-1}$ two sequences of QPSK coefficients $\{t_1[n]\}_{n=0}^{N/2-1}$ forms a pseudonoise (PN) sequence which is transmitted on the even subcarriers for the first training symbol
- *Zeros are transmitted on the odd subcarriers for the first training symbol*

  $\Longrightarrow$ the two halves of the first training symbol are identical
- The even coefficients for the second training symbol are

$$t_2[2n]t_1^*[2n] = t_3[2n]$$

  - A PN sequence transmitted on the even subcarriers to help determine the integer frequency offset
  - Odd subcarriers could contain other training data

Rewriting and leveraging the cyclic shift property

$$\epsilon = \frac{2q}{N} + \epsilon_{\text{frac}}$$

integer offset

fractional offset

$$\begin{aligned} e^{\mathrm{j}2\pi\epsilon N/2} &= e^{\mathrm{j}2\pi\left(\frac{2q}{N}+\epsilon_{\text{frac}}\right)\frac{N}{2}} \\ &= e^{\mathrm{j}2\pi\left(\frac{qN2}{N2}+\epsilon_{\text{frac}}\right)\frac{N}{2}} \\ &= e^{\mathrm{j}2\pi\epsilon_{\text{frac}}\frac{N}{2}}, \end{aligned}$$

- For a given symbol, with a larger frequency offset, after fine offset correction

$$y[n] = e^{\mathrm{j}2\pi\frac{2q}{N}n}\sum_{\ell=0}^{L} h[\ell]w[n-\ell] + v[n]$$

- In the frequency domain

$$\mathsf{y}[k] = e^{-\mathrm{j}2\pi\frac{2q}{N}L_{\mathrm{c}}}\mathsf{h}[((k-2q))_N]\mathsf{s}[((k-2q))_N] + \tilde{v}[k]$$

Estimating the integer ambiguity

- Let $y_1[k]$ and $y_2[k]$ the DFTs of the first and second received training symbols after fine offset correction

$$y_1[k] = e^{-j2\pi \frac{2q}{N} L_c} h[((k-2q))_N] t_1[((k-2q))_N] + \tilde{v}[k]$$

$$y_2[k] = e^{-j2\pi \frac{2q}{N} L_c} h[((k-2q))_N] t_2[((k-2q))_N] + \tilde{v}[k+N+L_c]$$

- Consider the product

$$y_2[2k] y_1^*[2k] = |h[((2k-2q))_N]|^2 t_2[((2k-2q))_N] t_1^*[((2k-2q))_N] + \tilde{v}'[k]$$

exploit differential encoding

$$= |h[((k-2q))_N]|^2 t_3[((2k-2q))_N] + \tilde{v}'[k]$$

Cross terms plus noise

- Solve LS problem

$$\hat{q} = \max_{p=0,1,\ldots,N/2-1} \frac{\left| \sum_{k=0}^{N/2-1} y_2[2k+2p] y_1^*[2k+2p] t_3^*[2k+2p] \right|^2}{\left(\sum_{k=0}^{N/2-1} |y_2[2k]|^2 \right)^2}$$

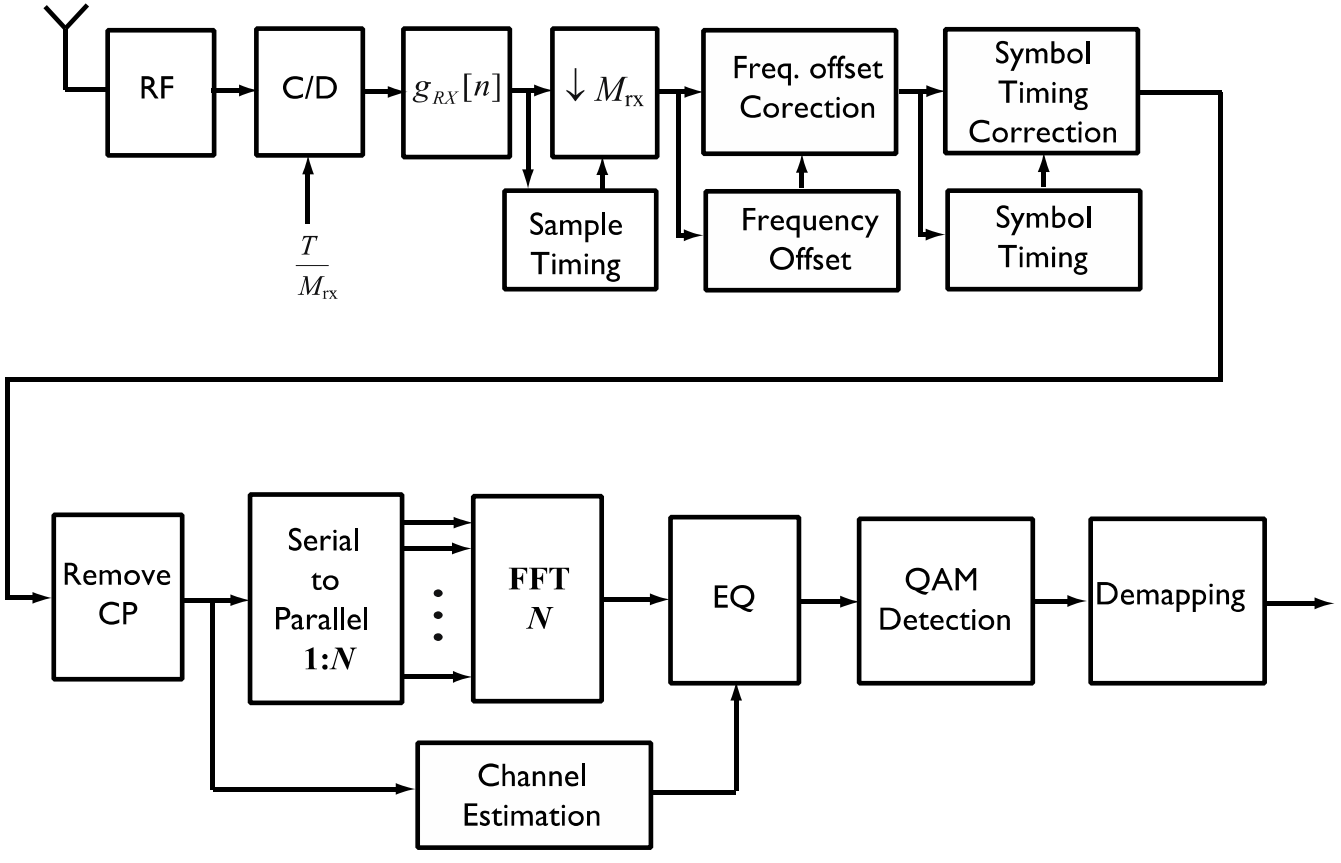
Schmidl-Cox algorithm

- The received samples can be corrected by $e^{-\mathrm{j}2\pi \frac{2\hat{q}}{N} n}$
- The final frequency offset estimate is

$$\epsilon = \frac{2\hat{q}}{N} + \hat{\epsilon}_{\text{frac}}$$

- A large range of offsets can be corrected

OFDM receiver with frequency offset correction



Frequency offset estimation in OFDM

Learning objectives

- Describe and implement carrier frequency offset estimation for an OFDM system